Day 3 - Notes

21 October 2026 07:20

Recap on Module 3 - Development & TRG → 1hr

Module 4 - Deployment & Inference options in SageMaker. → [Lab 9 & 10] → 1.5 hrs

Module 5 - Monitoring (Demo - No Lab) → 30 mins

↳ Publicly available "AWS workshops" (AWS own A/C)

Module 6 = updated with latest content

"Capstone Project" → 3 to 4 HRs.

Recap on Module #3: Model Development & TRG.

Amazon SageMaker AI TRG Jobs.

↳ To execute Model TRG using managed compute resources without having to manually provision

↑

TRG data, Algo | TRG code, Compute Config, Hyperparameters → output

↓

Model Artifacts.

3 Model development approaches:

① Built-in Algos

② Bring your own script (BYOS) → Managed Compute Resource + Framework

③ Bring your own Container

Parameters vs Hyperparameters

↑ learned by the Model

↑ Configuration values that control the Trg process

the Model during TRG

Trg process

⇒ Model Evaluation – Experiments, Debugger, Hyperparameter Tuning.

① SageMaker Experiments (MLFlow)
↳ Track runs, parameters, metrics, artifacts, results

② SageMaker Debugger → Help investigate issues during TRG.

③ Bias → SageMaker Clarify?
↳ even if a model performs well is it making predictions fairly / Bias?
↳ Supports Bias analysis & model explainability

✓ Data Prep

Prepare reliable, reusable datasets & features

① Canvas → Data Wrangler
↳ Low code / Not in millions
② EMR Cluster
③ SageMaker Processing Jobs
④ Feature Store
↳ [Feature groups]
↓
feature

✓ Model Develop / TRG.

TRAIN, evaluate, Tune, Troubleshoot & access our model for Bias

① Train = 'TRG Jobs'
② evaluate = SageMaker Experiments
↓
MLFlow
③ Hyperparameter Tuning
[0 – 2]
④ Troubleshoot = SageMaker Debugger
⑤ Bias = SageMaker Clarify

Module 4: Deployment & Inference

Module 4: Deployment & Inference

(1) SageMaker Model Registry → manage Model versions & approvals

(2) SageMaker pipelines → Automating Model-building workflow

- notebooks
- SageMaker AI

(3) Inference compare → real-time, Batch, Async, Serverless

endpoint config, scaling.

Module 5 = Monitoring

(1) SageMaker Model Monitor

(2) Amazon Cloudwatch & SNS.

Model Registry

✓ Manages & versions model packages ✓

Model cards

✓ Documents – model purpose, evaluation, risks, governance-related info ✓

Lab 9: SageMaker AI Pipelines & Model Registry.

Business use case: Customer churn prediction.

3 MLOps concepts

(1) Automation → SageMaker pipeline → ML jobs & their dependencies

(1) Automation → SageMaker pipeline → ... dependencies

(2) Quality control → Model AUC score greater than 0.75

(3) Governance → Model Registry

Complete Architecture of Lab

Customer Data

CSV files → SageMaker feature Store

↓

(1) Processing (Develop & TRG)

Prepare TRG, Validation, Test, Batch dataset (TRG, Validate, Test)

↓

(2) Hyperparameter Tuning

Train XGBoost candidate & select the Best Model

↓

(3) Model evaluation

Evaluate the Best Model — Test DS

↓

(4) Is Test AUC greater than 0.75

Yes

(5) Create Model

↓

(6) Batch Transform

No

STOP this Branch

↓

① Register the Model

↓

② Manual Approval

Real-time Inference

Model Artifacts + Container → SageMaker Model →
Prod Variant → Endpoint Config → Endpoint

Client App → Invoke Endpoint → Prediction.

Real-time = Provisioned Capacity → Predictable avail

Serverless : Req arrive → Capacity auto (of req) → Inference
↓
Autoscales

(9-5)
↓

Model + Sample Payloads + workload
↓
Inference Recommender.

Req < 1000 p/sec
100 ms

Real-time endpoint → provisioned server.
Batch Transform — job-oriented
Async → queue req & write result
Serverless → automatically provision.

Real-time

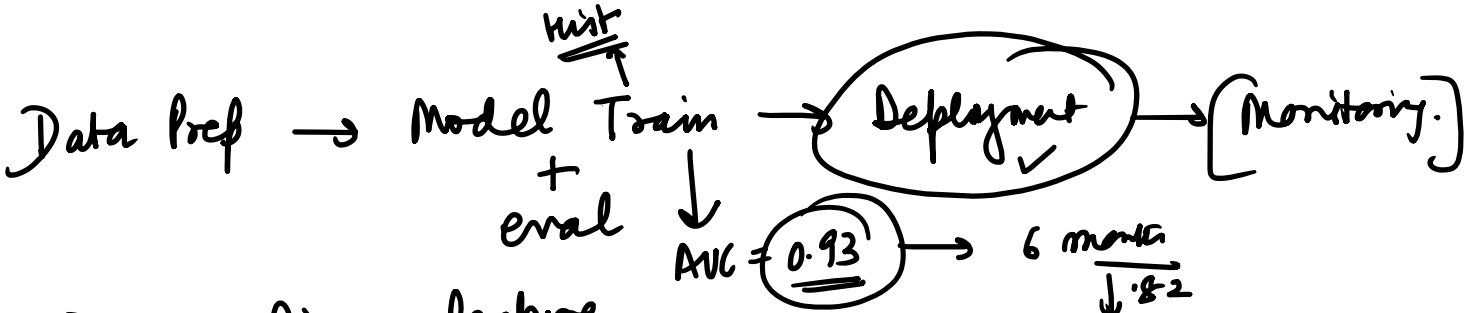
(1) Env Setup → load lib, inspect data, register the model
(2) Create endpoint → configure compute, enable data capture, deploy model
(3) Invoke endpoint → send sample records & retrieve predictions

Serverless

(1) Env Setup = Install dep, Configure AWS, upload & register the model
(2) Create Serverless endpoint = Configure memory & concurrency, then deploy
(3) Invoke endpoint = Send data & obtain predictions.

"Async"

(1) Env Setup = Install dep/lib, upload the model, upload input data
[(2) Create endpoint] = output location - AWS Roles.



Data, Bias, feature

(1) Data quality
(2) Model quality
(3) Modal Bias
(4) Feature Attribution

Model Monitor

(3) Modal Bias
(4) Feature Attribution

Data Drift · Income (TRG DS) = $75K
Prod Req = $45K

Concept Drift

Bias & Feature attribution

Demo: Monitoring.

End-to-end ML workflow

Model TRG → Prod Monitoring
using XGBoost
ML flow → experiments, Model Registry
SageMaker Pipelines, Amazon SNS

Business Problem

Bank wants to predict which customers are likely to subscribed to a term deposit.

Bank — info about customers:

(1) Age, occupation, marital status — 30-45
(2) Existing loans, financial
(3) Previous marketing interactions
indicators

(3) Previous marketing interactions
(4) Campaign details & Economic Indicators

ML Model — predicts
- 0 ⇒ unlikely to subscribe
- 1 ⇒ likely to subscribe

↓

Prod: 55-70

Overall Notebook:

Bank MkTG Dataset

TRG, validation, Test, reference data

↓

SageMaker XGBoost TRG

TRG results & metrics track in MLflow

↓

Real-time SageMaker Endpoint

Predict customer subscription & Capture req/resp

- Baseline Data in S3
  - TRG feature distribution
- Capture Data in S3
  - New Inference Traffic

↓

Monitoring

Detect feature drift & evaluate Model Quality

- MLflow
- Amazon SNS

ML flow

Reports, metrics, History

Amazon SNS

Conditional drift / Alerts